Vaaheesan S

Backend Software Developer

Python • FastAPI • Payments & APIs • React

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Portfolio GitHub LinkedIn LeetCode HackerRank SkillRack

Professional Experience

Backend Lead Intern – Hexaware Technologies *Feb 2026 – Aug 2026*

- Designed and implemented **RESTful APIs** and schema-driven backend services using Python and FastAPI for applications supporting **3,000 concurrent users**; led a 3-member backend sub-team within a 12-member engineering team.
- Developed **asynchronous** evaluation pipelines using Redis queues and `asyncio`, integrated Alembic for version-controlled database migrations, and enforced code quality using Ruff.
- Contributed to deployment validation, CI/CD workflows, **production monitoring**, and backend optimization, reducing application performance overhead by approximately 3%.

Full-Stack Developer Intern – GameChanger4u *May 2026 – Aug 2026*

- Integrated the **Razorpay payment gateway** for turf slot bookings end-to-end, implementing order creation, a custom React checkout flow, and **server-side HMAC-SHA256 signature verification** to prevent payment tampering.
- Developed **FastAPI microservices** with SQLAlchemy and Alembic, implementing **JWT authentication** and tenant header validation across a multi-tenant SaaS platform deployed on AWS ECS Fargate.
- Built frontend features for Gym and Academy modules using Next.js 14 (App Router), TypeScript, and Tailwind CSS, including member dashboards, slot booking flows, and analytics views.

Projects

RecoverAI – Agentic Payment Recovery & Safety Orchestration

Next.js, FastAPI, PostgreSQL, Redis, LangGraph, OpenAI/Gemini

- Built a **multi-agent payment recovery system** using LangGraph, coordinating Transaction Analyst, Customer Risk, and Strategy Selector agents to analyze payment failures, customer history, and risk factors.
- Engineered a deterministic **ActionGuard** validating AI actions against merchant limits, retry thresholds, and risk constraints, with Redis workers, PostgreSQL persistence, exponential backoff, authorization tokens, human escalation, idempotency, and audit trails.

GameChanger4u – Multi-Tenant Sports & Gym SaaS Platform

Next.js 14, TypeScript, FastAPI, PostgreSQL, Redis, AWS

- Architected a **multi-tenant microservices platform** using FastAPI, PostgreSQL, SQLAlchemy, and Alembic across authentication, academy, gym, billing, turf, and notification services.
- Implemented **tenant isolation** with AWS ECS Fargate and Terraform, alongside JWT authentication, tenant validation, Redis caching, idempotency middleware, and database migrations.

Telu Telecom Triage – AI-Powered Telecom Complaint Intelligence

Next.js, TypeScript, FastAPI, PostgreSQL, pgvector, Redis, ML/LLM

- Developed a **hybrid ML/LLM complaint triage pipeline** using TF-IDF + Logistic Regression with confidence-based routing to HuggingFace Llama-3.1-8B for complex complaints.
- Implemented **technical feature extraction, normalization, and complexity scoring** across six telecom dimensions, with PostgreSQL/pgvector, Redis, JWT/OTP authentication, and Docker orchestration.

Technical Expertise

Languages: Python, Java, SQL, JavaScript, TypeScript

Backend: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Alembic

Frontend: Next.js, React.js, Tailwind CSS

Data: PostgreSQL, MySQL, MongoDB, Redis

AI/ML: LangGraph, LLMs, pgvector, TF-IDF, Logistic Regression

Cloud & DevOps: AWS, Docker, Terraform, CI/CD, Git, GitHub

Competitive Programming

- **LeetCode** – Solved 131+ coding problems with a Contest Rating of **1655** (Top **17.5%**).
- **SkillRack** – Solved 308+ programming problems, Ranked #1 in College Programming Test Leaderboard, earned certifications in Data Structures (C) and SQL Basics.

Achievements

- Selected for the MSME 5.0 Government Funding Scheme for the project **MindSync+** (2026)
- 1st Place – QR Debugging Event (Namma VEC 2026)
- 2nd Place – App Development (Namma VEC 2025)

Education

B.E. Computer Science and Engineering – Velammal Engineering College *2023 – 2027*

CGPA: **8.37** | AISSCE (Class XII): 68.6% | AISSE (Class X): 81.0%